EXPERIMENT	IMPLEMENTATION OF PIG LATIN SCRIPTS FOR MANIPULATION if DATABASE	ROLL NO	23DI16	
		DAY	MONTH	YEAR
5				

AIM:

To implement of pig latin scripts for manipulation if database.

COMPONENTS REQUIRED:

S.no	FACILITIES REQUIRED	QUANTITY
1	System	01
2	Operating System	Ubuntu

PROCEDURE:

Step 1: Switch to the Hadoop User.

$~ su hdoop

```
dit@dit-OptiPlex-3060:~$ su hdoop
Password:
```

Step 2 : Start Hadoop Distributed File System (HDFS).

$~ start-dfs.sh

```
hdoop@dit-OptiPlex-3060:/home/dit$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [dit-OptiPlex-3060]
```

Step 3 : Start YARN Resources

$~ start-yarn.sh

```
hdoop@dit-OptiPlex-3060:/home/dit$ start-yarn.sh
Starting resourcemanager
WARNING: /home/dit/hadoop-3.2.3//logs does not exist. Creating.
mkdir: cannot create directory '/home/dit': Permission denied
ERROR: Unable to create /home/dit/hadoop-3.2.3//logs. Aborting.
Starting nodemanagers
```

Step 4 : Launch Apache Pig

$~ pig

```
hdoop@dit-OptiPlex-3060:/home/dit$ pig
2026-01-21 19:03:42,535 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-01-21 19:03:42,536 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-01-21 19:03:42,536 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecT
ype
2026-01-21 19:03:42,744 [main] INFO  org.apache.pig.Main - Apache Pig version 0.
16.0 (r1746530) compiled Jun 01 2016, 23:10:49
2026-01-21 19:03:42,744 [main] INFO  org.apache.pig.Main - Logging error message
s to: /tmp/hsperfdata_hdoop/pig_1769002422737.log
2026-01-21 19:03:42,793 [main] INFO  org.apache.pig.impl.util.Utils - Default bo
otup file /home/hdoop/.pigbootup not found
2026-01-21 19:03:42,988 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr
```

Step 5 : Execute Pig Latin Commands

grunt> A = load 'passwd' using PigStorage(':');
grunt> B = foreach A generate $0 as id;
grunt> dump B;

```
grunt> A = load 'passwd' using PigStorage(':');
2026-01-21 19:05:32,994 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - fs.default.name is deprecated. Instead, use fs.defaultFS
grunt> B = foreach A generate $0 as id;
grunt> dump B;
```

Step 6 : Review Execution Results.

```
Input(s):
Failed to read data from "file:///home/hdoop/passwd"

Output(s):
Failed to produce result in "file:/tmp/temp-971282463/tmp1160817182"

Counters:
Total records written : 0
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1362987935_0001
```

RESULT:

Thus, to implement of pig latin scripts for manipulation if database has been implemented successfully.

$~ su hdoop

```
dit@dit-OptiPlex-3060:~$ su hdoop
Password:
```

$~ start-dfs.sh

```
hdoop@dit-OptiPlex-3060:/home/dit$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [dit-OptiPlex-3060]
```

$~ start-yarn.sh

```
hdoop@dit-OptiPlex-3060:/home/dit$ start-yarn.sh
Starting resourcemanager
WARNING: /home/dit/hadoop-3.2.3//logs does not exist. Creating.
mkdir: cannot create directory '/home/dit': Permission denied
ERROR: Unable to create /home/dit/hadoop-3.2.3//logs. Aborting.
Starting nodemanagers
```

$~ pig

```
hdoop@dit-OptiPlex-3060:/home/dit$ pig
2026-01-21 19:03:42,535 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-01-21 19:03:42,536 INFO pig.ExecTypeProvider: Trying ExecType : MAPREDUCE
2026-01-21 19:03:42,536 INFO pig.ExecTypeProvider: Picked MAPREDUCE as the ExecT
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16.0 (r1746530) compiled Jun 01 2016, 23:10:49
2026-01-21 19:03:42,744 [main] INFO  org.apache.pig.Main - Logging error message
s to: /tmp/hsperfdata_hdoop/pig_1769002422737.log
2026-01-21 19:03:42,793 [main] INFO  org.apache.pig.impl.util.Utils - Default bo
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2026-01-21 19:03:42,988 [main] INFO  org.apache.hadoop.conf.Configuration.deprec
ation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.addr
```

grunt> A = load 'passwd' using PigStorage(':');
grunt> B = foreach A generate $0 as id;
grunt> dump B;

```
grunt> A = load 'passwd' using PigStorage(':');

Input(s):
Failed to read data from "file:///home/hdoop/passwd"

Output(s):
Failed to produce result in "file:/tmp/temp-971282463/tmp1160817182"

Counters:
Total records written : 0
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1362987935_0001
```

EXPERIMENT	WRITE A PIG LATIN SCRIPTS FOR FINDING TF-IDF VALUE FOR ANY DATABASE	ROLL NO	23DI16	
		DAY	MONTH	YEAR
6				

AIM:

To write a pig latin scripts for finding TF-IDF value for any database.

COMPONENTS REQUIRED:

S.no	FACILITIES REQUIRED	QUANTITY
1	System	01
2	Operating System	Ubuntu

PROCEDURE:

Step 1: Access the Hadoop User.

$~ su hdoop

```
dit@dit-OptiPlex-3060:~$ su hdoop
Password:
```

Step 2 : Create the Input File.

$~ nano student.txt

```
hdoop@dit-OptiPlex-3060:/home/dit$ cd
hdoop@dit-OptiPlex-3060:~$ nano ptc.txt
```

Step 3 : Verify File Creation

$~ ls

```
hdoop@dit-OptiPlex-3060:~$ ls
Desktop                 Pictures                pig_1767103639756.log   snap
dfsdata                 pig                     pig_1767105789409.log   student.txt
Documents               pig-0.15.0.tar.gz       pig_1767108576703.log   Templates
Downloads               pig-0.16.0.tar.gz       pig_1767108934723.log   tmpdata
filter_names.pig        pig-0.16.0.tar.gz.1     pig_1767108940059.log   Videos
hadoop-3.4.0            pig-0.16.0.tar.gz.2     pig_1767109475060.log   wget-log
hadoop-3.4.0.tar.gz     pig-0.16.0.tar.gz.3     pig_1769003290215.log   wordcount
hadoopdata              pig-0.17.0              pig_1769003614901.log   WordCount
Music                   pig-0.17.0.tar.gz       pig_1769004196179.log   WordCountLab
names1.txt              pig-0.17.0.tar.gz.1     ptc.txt
names.txt               pig-0.17.0.tar.gz.2     Public
```

Step 4 : Launch Apache Pig in Local Mode

$~ pig -x local

```
hdoop@dit-OptiPlex-3060:~$ pig -x local
2026-01-21 15:01:18,364 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-01-21 15:01:18,364 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-01-21 15:01:18,380 [main] INFO  org.apache.pig.Main - Apache Pig version 0.
16.0 (r1746530) compiled Jun 01 2016, 23:10:49
2026-01-21 15:01:18,380 [main] INFO  org.apache.pig.Main - Logging error message
s to: /home/hdoop/pig_1768987878379.log
2026-01-21 15:01:18,390 [main] INFO  org.apache.pig.impl.util.Utils - Default bo
```

Step 5 : Process the Data in the Grunt Shell

grunt> A = LOAD 'ptc.txt' USING PigStorage('\n') AS (line:chararray);

grunt> DUMP A;

```
grunt> A = LOAD 'student.txt' USING PigStorage(':');
2025-12-30 20:31:17,477 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation
 - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
grunt> B = FOREACH A GENERATE $0 AS name, $1 AS roll_number;
grunt> DUMP B;
2025-12-30 20:31:31,338 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation
```

Step 6 : View the Output.

```
nputFormat - Total input files to process : 1
2026-01-21 15:03:28,485 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
```

RESULT:

Thus, to write a pig latin scripts for finding TF-IDF value for any database. Has been implemented successfully.

$~ su hdoop

```
dit@dit-OptiPlex-3060:~$ su hdoop
Password:
```

$~ nano student.txt

```
hdoop@dit-OptiPlex-3060:/home/dit$ cd
hdoop@dit-OptiPlex-3060:~$ nano ptc.txt
```

```
  GNU nano 6.2                          ptc.txt
ayesha divya shreeja farhana kathija mithra
```

$~ ls

```
hdoop@dit-OptiPlex-3060:~$ ls
Desktop                 Pictures                pig_1767103639756.log   snap
dfsdata                 pig                     pig_1767105789409.log   student.txt
Documents               pig-0.15.0.tar.gz       pig_1767108576703.log   Templates
Downloads               pig-0.16.0.tar.gz       pig_1767108934723.log   tmpdata
filter_names.pig        pig-0.16.0.tar.gz.1     pig_1767108940059.log   Videos
hadoop-3.4.0            pig-0.16.0.tar.gz.2     pig_1767109475060.log   wget-log
hadoop-3.4.0.tar.gz     pig-0.16.0.tar.gz.3     pig_1769003290215.log   wordcount
hadoopdata              pig-0.17.0              pig_1769003614901.log   WordCount
Music                   pig-0.17.0.tar.gz       pig_1769004196179.log   WordCountLab
names1.txt              pig-0.17.0.tar.gz.1     ptc.txt
names.txt               pig-0.17.0.tar.gz.2     Public
```

$~ pig -x local

```
hdoop@dit-OptiPlex-3060:~$ pig -x local
2026-01-21 15:01:18,364 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
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16.0 (r1746530) compiled Jun 01 2016, 23:10:49
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```

grunt> A = LOAD 'ptc.txt' USING PigStorage('\n') AS (line:chararray);

grunt> DUMP A;

```
grunt> A = LOAD 'student.txt' USING PigStorage(':');
2025-12-30 20:31:17,477 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation
 - io.bytes.per.checksum is deprecated. Instead, use dfs.bytes-per-checksum
grunt> B = FOREACH A GENERATE $0 AS name, $1 AS roll_number;
grunt> DUMP B;
2025-12-30 20:31:31,338 [main] INFO  org.apache.hadoop.conf.Configuration.deprecation
```

```
nputFormat - Total input files to process : 1
2026-01-21 15:03:28,485 [main] INFO  org.apache.pig.backend.hadoop.executionengi
ne.util.MapRedUtil - Total input paths to process : 1
```